

Hariharan Gopal

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EDUCATION

Columbia University, Manhattan, NY

Graduating Dec 2026

MS in Electrical Engineering, Specialization VLSI

Relevant Coursework: VLSI Design, Advanced Digital Design, Analog-Digital Circuits, Digital Systems

University of Illinois at Urbana-Champaign, Urbana-Champaign, IL

Dec 2024

BS in Electrical Engineering with Honors

SKILLS

EDA/FPGA Tools: Cadence Virtuoso, Cadence Innovus, Cadence Conformal, Synopsys PrimeTime, Mentor Calibre, ModelSim, Vivado, Quartus, Spectre, UltraSim, KiCad

Physical Design Flow: RTL-to-GDSII, synthesis, floorplanning, PnR, CTS/post-CTS optimization, STA (corners/margins/derates), DRC/LVS signoffs, LEC verification, PI/SI analysis, PPA trade-offs, DFT (Scan, LBIST)

Languages: Python, C, C++, Java, SystemVerilog, Verilog, Tcl

PROFESSIONAL EXPERIENCE

Ayar Labs | Post-Silicon Validation Intern

May 2026 - Present

- Proposed a time-interleaved multi-ADC sampling scheme, cutting a 15.8 ms hardware-limited conversion rate to ~1 ms (16x speedup) for finer capture of thermal transients in ring resonator characterization.
- Analyze polarization-dependent loss and effective responsivity of the TeraPHY Rx subsystem, delivering validated metrics ahead of tapeout.
- Built a new Rx test station and calibration infrastructure enabling repeatable optoelectronic testing and TDECQ characterization in 500Gb/s+ SerDes, providing feedback to the design team directly for tapeout.

Tibara Inc. | Board Development and RTL Intern

May 2023 - Aug 2023

- Implemented peripheral RTL (SPI/I2C communication and debug functions) in C/SystemVerilog, driving a 75% efficiency boost in future testing.
- Redesigned Magnetron circuit board schematics/layouts in Eagle with long-line impedance matching and signal integrity analysis; modifications approved by the CTO for final production.

PROJECTS

1024-Point FFT Engine Design & Layout (TSMCN65)

Jan 2026 - May 2026

- Architected a 1024-point radix-2 FFT engine in Verilog with ping-pong SRAM banks and a butterfly datapath; taken through synthesis, PnR, and DRC/LVS-clean sign-off in Innovus for tapeout.
- Closed all setup/hold violations via added register staging, verified with PrimeTime STA, reaching 208.7 MHz post-APR (4.64 MSamples/s) at 0.41 mm², 5.49 pJ/sample.
- Validated spectral accuracy across 10,240 samples: 70.6 dB SQNR, sub-0.15% NRMSE per frame.

High-Efficiency 8-bit Processor ASIC Implementation (TSMCN65)

Aug 2025 - Dec 2025

- Designed and laid out an 8-bit accumulator-based microcontroller core (6-bit instruction set single-cycle execute) in Cadence Virtuoso using the TSMCN65 process design kit, integrating datapath blocks (MAC, multiplexer, shifter, bus driver) and opcode-driven control/decoder logic, exceeding the target frequency of 112MHz.
- Verified performance using Spectre and PrimeTime, running critical-path simulations and signoff STA to validate clock cycle frequency and confirm instruction-level correctness at transistor level, ready for tapeout.
- Improved efficiency and reduced area by applying a bitslice floorplan (uniform-height datapath) and a balanced phi1/phi2 clock buffer tree to handle heavy clock loading while minimizing skew/imbalance.

64-Tap FIR Filter RTL Design (TSMCN65)

Aug 2025 - Dec 2025

- Designed a compact 64-tap 16-bit fixed-point FIR filter accelerator (Q1.15 I/O) for 10 kS/s throughput, with a streamlined MAC-based ALU, register file, coefficient memory, and FSM controller.
- Reached 242 MHz post-synthesis Fmax within a 0.162 mm² footprint at 52.5 pJ/sample energy efficiency.
- Benchmarked time-domain output fidelity against a MATLAB fixed-point golden model, confirming sub-0.018% NRMSE across test vectors.